DOORS Complete Functional User and Administrator Manual

Data Orchestration and On Demand Reporting System

All features role based procedures security controls troubleshooting and operations

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Intended readers	HOG, administrators, Data Managers, Security Administrators, developers, reviewers, Data Viewers, External Users, API Users, GePNIC DBAs, application owners, operations and auditors
Scope	Dashboard, requests, query governance, reports, Agents, read APIs, write APIs, documents, subscriptions, categories, templates, PKI, users, logs, AIra, deployment and incident recovery
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Document Control

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1.0	25 October 2026	DOORS project team	Initial complete role-based operating manual	Audit Draft

How to Use This Manual

- Start with Section 1 and the chapter for your assigned role.
- Follow screen labels exactly; menus not permitted for your role will not be displayed.
- Do not share passwords, TOTP codes, private keys, PKCS12 files or wallet passwords.
- For every failure, retain the HTTP status, application code, time and DOORS trace ID.
- Actions described as approval, activation, permission, deployment or revocation require the authorised role and must not be performed using another person's account.

Contents

1 Quick Start and Role Map

2 Sign In MFA Session and Account Use

3 Navigation and Common Screen Behaviour

4 External Data Request

5 Read Query Submission and Approval

6 Reports and Exports

7 Agent Registration and Verification

8 Read API Client Onboarding

9 Write Service Proposal and Approval

10 DBA Installation Agent Mapping and Verification

11 Write API Client Onboarding and Testing

12 Write Record Outcomes and Versioning

13 Document Services

14 Subscriptions Access Windows and Usage

15 Execution Logs Troubleshooting and Incidents

16 AIra Assistant

17 User Security and Platform Administration

18 Routine Operations and Release Checks

19 Error Messages and Recovery

20 Role Checklists and Escalation

- Appendix A Field and Status Reference
- Appendix B Security Do and Do Not Guide
- Appendix C Glossary

1 Quick Start and Role Map

1.1 What DOORS Does

DOORS allows authorised people and applications to request, approve, read, write and download GePNIC data through governed services. Consumers do not receive database credentials or unrestricted SQL access. Every material action is restricted by role and scope and leaves traceable evidence.

1.2 Request Planes

Plane	Use it for	Typical users	Do not use it for
Admin	Governance, registries, approvals, Agents, clients, policies, audit and administration	Administrator, Security Administrator, Data Manager, Developer	External self-service or unattended API calls
External	Data-request submission, request status and approved external reports	External User, Data Viewer	Governance administration or gateway execution
API	Machine-to-machine read, write and document operations	Registered applications and SDKs	Interactive portal sign-in

1.3 Role to Task Map

Role	Primary tasks	Main menu areas	Cannot do
External User	Submit and track data requests	Request Submission, My Requests	Approve requests or access unapproved reports
Developer	Prepare and test SELECT query proposals	Request Submission, Query workflow	Approve own SQL
Data Viewer	Run approved reports and permitted exports	Reports Hub	See or change SQL
Data Manager	Approve requests/queries/write contracts; manage Agents, clients and permissions; investigate logs	Governance Hub, Reports Hub, Documents Hub, Execution Logs	Self-approve own governed proposal
Security Administrator	Users, roles, VPN/IP, MFA recovery and security review	Administration and security screens	Use security administration to bypass data governance
API User	View assigned integrations, SDKs, notices and usage	API portal or assigned client view	Administer other clients
GePNIC DBA	Run approved SQL and validate database privileges	Database console outside DOORS; Write Library SQL	Give runtime user DDL or unrestricted table writes
Application Owner	Protect keys, configure SDK/application and test APIs	Generated SDK, OpenAPI, Swagger	Share private keys or reuse request IDs with changed payload
Auditor	Review approvals, configurations, logs and evidence	Read-only evidence access	Change operational state

1.4 Before You Begin

- Confirm the correct environment and URL before entering credentials.
- Use a supported current browser and correct system time for MFA and request signing.
- Connect through the required NIC/VPN network where policy applies.
- Ensure your current IP is included in your user or client allowlist.
- Know the Agent/service/client names assigned to your work.
- Application owners must have Java 21 where using generated Java SDKs.

2 Sign In MFA Session and Account Use

2.1 First Sign In

Activate a newly approved human account.

Step	What to do	What you should see	Important control or tip
1	Open the assigned Admin or External login URL.	Login form and CAPTCHA are visible.	A 421 response normally means the wrong plane/hostname.
2	Enter username/email, temporary password and CAPTCHA; select Sign In.	Password step succeeds or an MFA/enrolment step appears.	Do not repeatedly retry an unknown password; use approved recovery.
3	Change the temporary password when prompted.	Password accepted and reset flag cleared.	Follow the displayed password rules; never reuse a shared password.
4	Enrol TOTP MFA using your approved authenticator and confirm the six-digit code.	MFA enabled and dashboard opens.	Store recovery information according to security policy.
5	Verify your displayed account name, role and environment.	Correct role-specific dashboard/menu.	Report wrong role or organisation immediately.

2.2 Normal Sign In and Logout

Use an existing account safely.

Step	What to do	What you should see	Important control or tip
1	Open the correct URL and verify HTTPS and hostname.	DOORS login page.	Do not proceed through certificate warnings.
2	Enter password and CAPTCHA.	MFA challenge.	Passwords are never sent in chat/email.
3	Enter the current TOTP code.	Role dashboard.	If codes fail, verify device time before requesting reset.
4	Complete work; use the account menu and select Logout.	Login page; session is invalidated.	Close shared-browser windows after logout.

2.3 Password or MFA Recovery

Situation	User action	Security Administrator action	Expected result
Forgotten password	Contact authorised support; provide identity and no secret	Verify identity, reset credential and revoke sessions	Forced password change at next login
Lost MFA device	Request formal MFA recovery	Verify authority, reset MFA and revoke sessions	New TOTP enrolment
Current IP not whitelisted	Provide current official source IP and reason	Append approved CIDR/IP without deleting valid existing entries	Login allowed only from approved sources
Account inactive/rejected	Contact approving authority	Review status and audit history; activate only after approval	No silent bypass

3 Navigation and Common Screen Behaviour

UI element	Meaning	How to use it
Roadmap	Stages of the current workflow	Use the highlighted stage for the next required action
Red asterisk	Mandatory field	Complete before preview, test or submit
Status badge	Current governed state	Hover or open details for verification/error information
Refresh	Reload server state	Use after another user approves or an external deployment completes
Review and Approve	Open a pending governed proposal	Reviewer must inspect content and rerun tests after material edits
View Generated SQL	Inspect approved/proposed DBA objects	Review; do not execute unless approved and hash verified
Credentials and Security	API key, public key, IP and activation	Complete before machine testing
Service mappings	Client-to-service and Agent route	Mapping alone is not database permission
Service permission	Database service/instance grant	Apply approved SQL and confirm enabled state
Versions and SDKs	Guides, OpenAPI and UAT/Production packages	Download only the integration assigned to the client
Trace ID	Cross-service diagnostic identifier	Record it for troubleshooting

3.1 Safe Form Use

- Read the current roadmap stage before entering data.
- Use Preview or Analyse before submitting where offered.
- A disabled button normally means a required field or prerequisite is incomplete; read the nearby guidance.
- Opening a record uses a modal or drawer. Complete or close it before working on another row.
- Do not use browser Back during a multi-stage approval or upload operation unless the page instructs you to do so.

3.2 Dashboard and Work Queues

3.2.1 Use the Dashboard

Identify work requiring attention without changing governed state.

Step	What to do	What you should see	Important control or tip
1	Confirm the displayed mode, account and environment.	Role-correct dashboard and menu.	Stop if the account, role or environment is wrong.
2	Review request, approval, Agent, API and failure counters.	Counts and status cards for authorised scope.	A zero or stale count must be confirmed from the relevant queue.
3	Open a card or the corresponding menu item.	Filtered queue or operational screen.	Dashboard cards are navigation aids, not approval evidence.
4	Use Refresh after another user or external deployment completes.	Current server state.	Avoid repeated refresh loops during long operations.

3.3 Complete Feature Map

Area	What it manages	Primary users	Completion evidence
Request Submission	Business data needs and governed SQL or JSON proposals	External User, Developer, Data Manager	Request or proposal ID and status
Governance Hub	Independent approval, Query Library, Remote Agents and Write Library	Data Manager, Reviewer	Decision history and approved version
Reports Hub	Execution of approved read services and controlled export	Data Viewer, Data Manager	Result, row count and trace ID
Documents Hub	Eligible document discovery and integrity checked download	Authorised users and API clients	Checksum and receipt
API Client Registry	Read and write client credentials, mappings, permissions, versions and SDKs	Data Manager, Security Administrator, Application Owner	Active client plus explicit service scopes
API Subscriptions and Usage	Access periods, schedules, notices, service state and usage	Data Manager, API administrator	Effective entitlement and usage record
Template Contracts	Reusable approved request or query structures	Data Manager, Developer	Published template version
Category Master	Controlled classification values used by services	Administrator, Data Manager	Active category and subcategory
PKI Certificate Management	Public certificates, fingerprints and validity	Security Administrator	Verified active certificate
Execution Logs	End-to-end audit and troubleshooting evidence	Data Manager, Operator, Auditor	Trace-correlated event chain
Documentation Hub and Alra	Published help, SOPs and grounded assistance	All authorised users	Relevant source or guided answer

4 External Data Request

4.1 Submit a Data Request

Request data with enough information for independent review.

Step	What to do	What you should see	Important control or tip
1	Sign in on the External Plane and open Request Submission.	New request form.	Your identity is taken from the session.
2	Enter a clear title, business purpose, fields, date range, frequency and justification.	Mandatory indicators clear.	Ask only for the minimum data needed.
3	Attach permitted supporting material where required.	Filename/type displayed.	Do not upload passwords, keys or unrelated personal data.
4	Review and submit.	Request ID and Pending status.	Save the request ID.
5	Open My Requests to track the decision and remarks.	Pending, Approved, Rejected or returned status.	A request approval does not automatically create an API/query.

4.2 Review a Data Request

Approve, reject or return a request as Data Manager.

Step	What to do	What you should see	Important control or tip
1	Open Governance Hub and pending data requests.	Queue with proposer, date and status.	Use your own authorised account.

Step	What to do	What you should see	Important control or tip
2	Open the request and verify purpose, minimisation and ownership.	Full request and attachment metadata.	Do not rely only on the title.
3	Enter governance remarks and choose the permitted decision.	Confirmation of state change.	Remarks must explain the decision.
4	Verify transaction history.	Proposer and reviewer events are visible.	History must not be replaced by only the latest approver.

4.3 Return Clarify and Resubmit a Request

Resolve incomplete requirements without losing transaction history.

Step	What to do	What you should see	Important control or tip
1	Reviewer returns the request and records each information gap.	Returned status and reviewer remarks.	Use rejection only when the request should not proceed.
2	Proposer opens My Requests and reads the remarks.	Editable returned request or revision action.	Do not create a duplicate request unless instructed.
3	Update scope, fields, dates, purpose or attachments and resubmit.	New history event and Pending status.	Remove obsolete or sensitive attachments.
4	Reviewer compares the revision with the earlier version and decides.	Complete proposer and reviewer trail.	Approval applies only to the current revision.

Request status	Meaning	Next action
Draft	Not submitted	Complete mandatory information
Pending	Awaiting authorised review	Wait or provide clarification
Returned	Correction required	Revise and resubmit
Approved	Business need accepted	Continue with the governed implementation
Rejected	Not permitted or justified	Read remarks; submit anew only if circumstances change
Closed	Workflow completed	Retain the request ID for audit

5 Read Query Submission and Approval

5.1 Create and Test a Query

Convert an approved need into a parameterised SELECT service.

Step	What to do	What you should see	Important control or tip
1	Open Request Submission and start a query proposal or continue the approved request.	Title, purpose and SQL editor.	Use a service name meaningful to consumers.
2	Enter one SELECT statement. Replace inputs with :p_ names such as :p_startDate.	Parameter inputs appear only for :p_ tokens.	Format text such as HH:MI is not an input parameter.

Step	What to do	What you should see	Important control or tip
3	Select a sandbox Agent and supply representative values.	Test-value fields and eligible sandbox Agent.	Use safe, bounded dates.
4	Select Run Dry-Run.	Columns, sample rows, time and result or exact rejection.	DDL/DML, stacked or unsafe SQL is rejected.
5	Correct SQL or values and rerun until successful.	Successful dry-run evidence.	A material SQL edit invalidates the earlier test.
6	Submit for approval.	Pending proposal under Request Approvals.	The proposer cannot approve it.

5.2 Review and Approve a Query

Independently verify SQL and authorised infrastructure.

Step	What to do	What you should see	Important control or tip
1	Sign in as an independent reviewer and open Request Approvals.	Pending write/read approval queues.	Do not use the proposer's account.
2	Select Review and Approve and inspect purpose, SQL and parameters.	SQL editor and test-value fields.	Only :p_ parameters should appear.
3	Edit SQL if necessary, select permitted Agents and rerun dry run.	Successful current test.	Changed SQL cannot be approved before retest.
4	Enter governance note and select Approve and Save or Reject.	Approved version or rejection status.	Explain material security/data decisions.
5	Open Query Library and confirm version, Agents and status.	Approved query entry.	Later SQL changes require a new governed revision.

5.3 Query Authoring Rules

Rule	Correct example	Avoid
Parameter prefix	:p_startDate, :p_endDate	:startDate or accidental :MI
SQL operation	SELECT with explicit joins	INSERT, UPDATE, DELETE, DDL or stacked statements
Dates	date(c.publishdate) BETWEEN CAST(:p_startDate AS date) AND CAST(:p_endDate AS date)	String concatenation of supplied values
Joins	Explicit JOIN ... ON	Unexplained Cartesian joins
Result size	Required columns and bounded date/filter	SELECT * over unbounded operational data
Formatting	to_char(ts,'DD-Mon-YYYY HH:MI PM')	Treating PostgreSQL format tokens as parameters

5.4 Create a New Query Revision

Change an approved read service without silently editing Production.

Step	What to do	What you should see	Important control or tip

Step	What to do	What you should see	Important control or tip
1	Open Query Library and create a revision of the approved query.	New draft populated from the approved definition.	The current approved version remains traceable.
2	Edit SQL, parameters, output metadata or purpose.	Draft revision with copied mappings.	Use only SELECT and :p_ parameter names.
3	Run a fresh dry run with safe bounded values.	Current columns, types, sample and timing.	Every SQL change invalidates earlier test evidence.
4	Submit for independent approval.	Pending revision linked to its predecessor.	The proposer cannot approve it.
5	After approval, confirm mappings and regression-test affected clients/reports.	New approved version and successful tests.	Do not assume old consumers accept changed columns.

5.5 Query Library Operations

Action	Purpose	Control
View definition	Inspect approved SQL, parameters and ownership	Check version and status
Map Agents	Select governed databases where the query may run	Agent health and authorisation
Disable	Stop new execution while retaining history	Record reason and impact
Create revision	Govern SQL, parameter or output changes	New dry run and approval
Catalogue	Maintain discoverability metadata	Must not hide an SQL change
History	Review proposal, tests and decisions	Chronological immutable evidence

6 Reports and Exports

6.1 Run an Approved Report

Run an approved query without accessing its SQL.

Step	What to do	What you should see	Important control or tip
1	Open Reports Hub and select an approved report.	Report description and parameter form.	Only reports allowed for your role are shown.
2	Enter values and select an eligible Agent if prompted.	Validated inputs.	Agent list is the intersection of query and user scope.
3	Run the report.	Bounded rows, columns and execution information.	Record the trace ID if the result is unexpected.
4	Sort/filter in the permitted UI and export if allowed.	CSV, Excel or PDF download.	Downloads are re-authorised and owned by your session.

6.2 Validate Results and Exports

Confirm that a report can be relied upon and shared.

Step	What to do	What you should see	Important control or tip
1	Check report, parameters, Agent, run time and row count.	Execution context beside the result.	Do not compare different instances as if identical.
2	Review empty, truncated or unusually large results.	Clear empty or result-limit message.	Change approved parameters, not SQL, from Reports Hub.
3	Select an allowed export format.	Downloaded file with expected columns.	Apply classification and retention policy.
4	Reconcile a small sample with the screen.	Matching values and encoding.	Report mismatches with trace ID and checksum.
5	Delete temporary exports when no longer required.	Local handling completed.	Avoid shared Downloads folders for sensitive data.

7 Agent Registration and Verification

7.1 Register an Agent

Make one GePNIC database/document node available for governed use.

Step	What to do	What you should see	Important control or tip
1	Confirm Agent API is deployed, HTTPS works, target profile exists and transit secret matches Master.	Agent health/governance endpoint responds.	Database passwords remain Agent-side.
2	Open Governance Hub, Remote Agents and choose register/add.	Agent form.	Use the exact Agent ID configured for routing.
3	Enter display name, HTTPS base URL, environment and metadata; mark sandbox only when intended.	Saved inactive/verification state.	One Agent represents one governed GePNIC database in the current design.
4	Run health/readiness checks.	Read and write readiness shown separately.	Read success does not imply write permission.
5	Activate only after successful checks and operational approval.	Active Agent.	Investigate route, TLS, secret, target and DB failures separately.

7.2 Common Agent Failures

Message	Likely cause	Check
Agent requires HTTPS	Non-HTTPS base URL outside permitted local development	Agent registry URL and reverse proxy TLS
HTTP 302	Authentication/redirect or incorrect proxy location	Direct endpoint must not redirect
HTTP 404	Wrong route prefix or old Agent build	Proxy mapping and supported Agent endpoint
HTTP 403	Source-IP/proxy or transit-secret denial	Apache/Nginx policy and matching dedicated secret
HTTP 422	Target/profile/schema/permission validation failed	Agent log exact diagnostic, target name, DB roles and function
HTTP 503	Agent unavailable or target connection failure	Service status, logs, JDBC connectivity

Message	Likely cause	Check
HTTP 504/result unknown	Timeout after dispatch	Trace logs and idempotent retry guidance

7.3 Modify Disable or Retire an Agent

Change infrastructure without orphaning governed services.

Step	What to do	What you should see	Important control or tip
1	Identify mapped queries, write services, documents, users and clients.	Impact list.	Do not disable first and investigate later.
2	Register and verify replacement capacity where required.	Healthy replacement Agent.	Read and write readiness are checked separately.
3	Move governed mappings through their authorised screens.	Updated mappings and audit events.	Database identity must remain unambiguous.
4	Disable the old Agent and run regression tests.	No new traffic to the retired Agent.	Retain historical logs and mappings.
5	Remove secrets and service access after retention and rollback periods.	Closed retirement record.	Do not delete audit evidence.

8 Read API Client Onboarding

8.1 Register and Authorise a Read Client

Prepare an external application for governed reads.

Step	What to do	What you should see	Important control or tip
1	Open API Client Registry and Read Services; create or select a client.	Client row with ID and action buttons.	Use one client identity per accountable application/environment.
2	Open credentials/security and securely record or rotate the API key.	Masked key and controlled reveal/copy.	Do not place keys in tickets or source control.
3	Register/analyse the application's public certificate and verify fingerprint.	Configured status and key details.	Production private key remains with the client.
4	Add approved source IP/CIDR values and activate the client after checks.	IP and active status.	Activation alone grants no data access.
5	Map approved Queries and Agents; add segregation where required.	Authorised mappings.	Runtime uses the intersection of client, query and Agent scopes.
6	Open Versions and SDKs and download the assigned UAT/Production package and checksum.	SDK/guide/OpenAPI files.	Verify SHA-256 before distribution.
7	Run UAT and confirm request in Execution Logs.	Successful response and trace.	Only then approve Production activation.

8.2 Configure Read Client Permissions

Apply every independent control required for a successful call.

Step	What to do	What you should see	Important control or tip
1	Confirm API key, public key, IP allowlist and client activation.	All credential indicators configured.	These identify and admit the caller; they do not grant data.
2	Map the client to approved queries and reports.	Query mappings.	Only explicitly mapped services are callable.
3	Map permitted Agents and segregation rules.	Agent and segregation entries.	Effective access is the intersection of all controls.
4	Confirm subscription, access period and schedule.	Effective current entitlement.	An expired period denies an otherwise valid client.
5	Download the correct environment package from Versions and SDKs and verify SHA-256.	Guide, SDK/OpenAPI and checksum.	Do not mix UAT and Production identities.
6	Run a bounded request and inspect the trace in Execution Logs.	Successful authorised response.	Never broaden access merely to make a test pass.

8.3 Direct Read API Responsibilities

- Use the API hostname and endpoint from the generated guide; the Admin portal URL is not a machine endpoint.
- Send the assigned API key and implement signing and encryption exactly as documented by the generated client.
- Use only mapped Agent IDs and :p_ parameters from the approved query.
- Validate TLS and server trust; diagnostic certificate workarounds are not Production designs.
- Use bounded retry and backoff for 429 and only retry operations that are safe.
- Record status, application code and trace ID without logging private keys, wallet passwords or AES keys.

9 Write Service Proposal and Approval

9.1 Propose a Write Service

Define how one JSON business document is stored and versioned.

Step	What to do	What you should see	Important control or tip
1	Open Request Submission and choose New Write JSON Proposal.	Write roadmap and Service Details.	Follow the highlighted roadmap stage.
2	Enter Service Title, lowercase Service Code and Purpose.	Valid fields; service code guidance.	Code uses lowercase letters/hyphens and no spaces.
3	Upload or paste representative JSON and select Analyse JSON.	Detected JSON fields.	Use representative structure, not secrets or real sensitive data.
4	Review each field, readable database column, required status and PostgreSQL type.	Editable mapping grid.	The full JSON is preserved in payload; extracted columns are optional projections.
5	Set varchar length/constraints where supported and select the Record Key fields.	Record-key indicator.	Choose stable fields that uniquely identify one business record.
6	Remove projections not needed and select Preview Tables and Rules.	Object/rule preview and View Generated SQL.	Removing a projection does not necessarily remove the field from payload acceptance.
7	Run Sandbox Write and Readback using an authorised sandbox Agent.	Passed evidence.	Sandbox leaves no persistent test objects/records.
8	Select Submit for Approval.	Pending proposal and contract version.	Save proposal/version ID.

9.2 Review a Write Contract

Independently validate JSON, database and consumer impact.

Step	What to do	What you should see	Important control or tip
1	Sign in as another authorised reviewer and open Request Approvals.	Pending write proposals.	Proposer cannot self-approve.
2	Select Review and Approve; compare purpose, JSON, key, required fields, types and columns.	Full definition and generated SQL.	Check compatibility with GePNIC consumer logic.
3	Edit permitted structure if necessary and rerun validation/sandbox.	Current successful evidence.	Material edits require retest.
4	Enter remarks and approve or reject.	Approved contract in Write Library.	Approval does not create database tables.
5	For later changes, choose Create New Revision from the approved service.	Copied fields/types/keys in a new proposal.	Do not re-enter or silently replace the approved contract.

10 DBA Installation Agent Mapping and Verification

10.1 Install Approved Write Objects

Create/migrate the approved structures in the mapped GePNIC database.

Step	What to do	What you should see	Important control or tip
1	In Write Library open the approved service and download/view DBA SQL.	SQL with explanatory notes and definition hash.	Use the latest approved contract version.
2	DBA compares the SQL definition hash with DOORS and reviews prerequisites, owners and grants.	Matching hash and approved change record.	Stop if hash differs.
3	Back up/export schema and privilege baseline; run SQL as designated DBA/schema owner.	Transaction commits without error.	Do not run as doors runtime user.
4	Confirm doors_write tables, latest view and store function and their owners.	Expected generic/service object names.	PUBLIC access must be revoked.
5	Return to Write Library, open Map and Verify, select one or more Agents and register mapping.	Mapped Agents with Verification pending.	Current UI uses Agent-only mapping; Agent corresponds to one GePNIC DB.
6	Run Verify Schema and Permissions separately for each Agent.	Verified badge or exact failing check.	Verification checks objects, columns, function owner/settings and grants.
7	Retain SQL, hash, execution output and verification evidence.	Auditable deployment package.	Do not edit generated SQL informally.

11 Write API Client Onboarding and Testing

11.1 Authorise a Write Client

Grant one application access to one verified write service.

Step	What to do	What you should see	Important control or tip

Step	What to do	What you should see	Important control or tip
1	Open API Client Registry, Write Services and select/create the intended client.	Write-services client table and roadmap.	Read and write permissions are independent.
2	Open Credentials and Security; configure API key, public key, source IPs and activation.	Configured indicators.	The Production private key remains with the application owner.
3	Open Service mappings and register the approved verified service/Agent mapping.	Disabled mapping initially.	Destination is fixed to preserve retry consistency.
4	Enable mapping after readiness review.	Active mapping.	Client activation alone is insufficient.
5	Open Service permission, choose Allow and generate/apply the database scope SQL.	Enabled client/service grant in doors_write.	This grants all record keys for this service/destination, not DDL.
6	Set the client-wise maximum changed versions where policy requires.	Version limit displayed/configured.	Identical JSON does not consume a version.
7	Open Versions and SDKs; choose UAT or Production package.	ZIP, SHA-256 and email text.	UAT bundles test identity; Production requires client identity.
8	Test with SDK or Swagger and verify stored data and logs.	STORED or UNCHANGED response.	Use approved sample data first.

11.2 Use the UAT Java SDK

Submit approved JSON with the generated client.

Step	What to do	What you should see	Important control or tip
1	Extract the ZIP and verify the provided SHA-256 checksum.	WriteClient.java, sample.json, wallet properties, identity/trust files and docs.	Read docs/index.html and README before running.
2	Open a terminal in the extracted directory.	Prompt shows SDK folder.	Java 21 must be available.
3	Review sample.json and wallet/trust settings; do not change service metadata embedded by the generator.	Valid sample and resolvable files.	Never email the bundled test password outside approved UAT distribution.
4	Run: java WriteClient.java sample.json	Request ID, Versioning: automatic, HTTP status and JSON response.	The SDK generates request ID and retains retry metadata.
5	Run the same unchanged JSON again.	HTTP 200 UNCHANGED or replay as applicable; no new version.	A changed JSON normally creates the next record version.
6	Search Execution Logs by request or trace and confirm destination Agent.	Decrypted authorised business context and outcome.	Cryptographic secrets are not logged.

11.3 Use Swagger Write Test

Perform a controlled browser test for an authorised client.

Step	What to do	What you should see	Important control or tip
1	Open Versions and SDKs, choose Swagger and select the intended write service.	Write-test roadmap.	Do not use an unrelated client/service.
2	Select the PKCS12/PFX identity and enter its key password.	Identity stage complete and filename retained.	The browser must not upload the private key before the explicit encrypted request step.

Step	What to do	What you should see	Important control or tip
3	Upload a JSON file or paste/edit the sample payload.	Payload preview.	Use the approved contract.
4	Select Encrypt/Prepare.	Encrypted request envelope and readiness state.	Signing covers the exact plaintext.
5	Select Post/Execute.	HTTP status and response JSON.	Do not reuse a request ID with changed JSON.
6	Record the response and verify backend/latest view if authorised.	Matching request ID and record version.	Close the test and clear sensitive local files if required.

12 Write Record Outcomes and Versioning

Outcome	Meaning	Record version effect	What the caller should do
STORED	Changed valid JSON stored	Increments automatically	Record requestId and returned version
UNCHANGED	Canonical JSON equals latest for the business key	No increment	Treat as success; no need to resubmit
replayed=true	Same request ID and same logical request already completed	No additional increment	Use returned original outcome
IDEMPOTENCY_CONFLICT	Same request ID reused with changed content	No write	Generate a new request ID for a genuinely changed submission
VERSION_LIMIT_REACHED	Configured changed-snapshot limit reached	No write	Contact DOORS Administrator; do not loop
TENDER_ACCESS_DENIED or access denied	Client/service/database scope not enabled	No write	Correct mapping and service grant
WRITE_RESULT_UNKNOWN	Master timed out after dispatch	Unknown until receipt checked	Retry identical payload using same request ID
Contract/type rejection	Required/key/type/unknown-field rule failed	No write	Correct JSON or create a governed contract revision

12.1 Contract Evolution

Change	Required action
Add or remove an extracted column	Create New Revision, analyse JSON, review all copied mappings, test, independently approve and apply DBA migration
Change PostgreSQL type/length/required flag	New governed revision plus data-compatibility review and DBA migration
Change record key	Treat as a high-impact revision; assess identity/version continuity and migration explicitly
New unknown JSON field during permissive rollout	Payload may be retained according to active unknown-field policy but is not automatically projected; create revision before consumers depend on it
Remove field from accepted JSON	Change contract required/allowed structure through revision; removing a projection alone is not enough
No structural change, new data only	Submit normally; automatic record versioning applies

13 Document Services

13.1 Download an Approved Document

Discover and retrieve a file without direct server-path access.

Step	What to do	What you should see	Important control or tip
1	Use the assigned API/SDK to run the approved manifest query.	Eligible document metadata and opaque identifier.	The caller does not supply arbitrary filesystem paths.
2	Request the document using its issued identifier.	Eligibility decision and stream response.	Policy checks run again at download time.
3	Allow the SDK to write a temporary part file and verify SHA-256.	Final filename appears only after checksum match.	Do not rename partial files as complete.
4	Allow/send the download receipt.	Correlation and source/destination checksums recorded.	Retain trace/correlation for support.
5	If denied, provide identifier, client, time, status/code and trace ID to Data Manager.	Auditable investigation.	Do not ask for direct server access as a workaround.

13.2 Configure a Document Service

Publish a controlled document retrieval service without exposing paths.

Step	What to do	What you should see	Important control or tip
1	Approve the business need and define manifest metadata and eligibility rules.	Governed document definition.	Use opaque identifiers, never user-supplied filesystem paths.
2	Map the approved query and document Agent.	Authorised discovery and content routes.	Metadata and bytes may use separate checks.
3	Configure permitted clients, users, file types, size/rate policies and access windows.	Explicit policy entries.	Read-query access alone does not grant document download.
4	Test eligible, ineligible, missing and checksum-failure cases.	Expected allow/deny responses and receipts.	Never bypass integrity checks for a test.
5	Publish guide/SDK and monitor downloads, failures and receipts.	Traceable operational evidence.	Usage metrics must not contain document content.

13.3 Document Download Troubleshooting

Symptom	Check	Safe action
Manifest returns no document	Query parameters, eligibility, mapped Agent	Correct authorised input or mapping
403 download denied	Client/user policy, IP, time window, identifier ownership	Fix the specific policy; do not share a direct path
404 document absent	Opaque ID, source record and Agent content root	Verify source synchronisation
Checksum mismatch	Source checksum, transfer, partial file and receipt	Delete partial output and investigate integrity
Slow or interrupted stream	Size/rate policy, network and Agent health	Resume only as the SDK permits
Duplicate download question	Receipt and correlation history	Use audit evidence rather than guessing

14 Subscriptions Access Windows and Usage

14.1 Configure API User Access

Manage machine-service availability independently from interactive reports.

Step	What to do	What you should see	Important control or tip
1	Open API Subscriptions and Usage and select the API user/client.	Current licence, Agent, status and version.	Do not edit another user's subscription.
2	Set service enabled/disabled and assigned Agent.	Updated current state.	Manual suspension denies applicable machine calls.
3	Create accounting/access periods with dates and amounts where used.	Versioned period history.	Periods are retained; do not destructively replace history.
4	Configure weekly slots and date-specific windows.	Calendar/schedule view.	Times use Asia/Kolkata unless explicitly stated.
5	Publish notices and review egress/usage metrics.	User-visible notice and byte/request data.	Usage stores metrics, not document/response content.

14.2 Rate Limits

Rate limits control request frequency and protect services from bursts and loops. They are separate from write record-version limits. HTTP 429 means the caller must wait for the allowed window, honour Retry-After where supplied and use bounded backoff. Repeated identical write JSON does not consume a record version, although it still consumes request capacity.

15 Execution Logs Troubleshooting and Incidents

15.1 Investigate a Request

Find the exact stage and reason for a failure.

Step	What to do	What you should see	Important control or tip
1	Collect timestamp, environment, endpoint, HTTP status, application code, request ID and trace ID.	Complete support facts.	Never request a password/private key.
2	Open Execution Logs and filter by trace ID first; otherwise use client, API key, Agent, endpoint or time.	Matching event chain.	Log viewing is itself audited.
3	Confirm caller/client, source IP, operation, approved parameters and destination Agent.	Readable authorised request context.	Private keys, passwords and AES keys remain excluded.
4	Identify the failure layer: UI/proxy, authentication, policy, Master, Agent, database or downstream timeout.	Specific stage/diagnostic.	Do not change multiple configurations at once.
5	Apply the least disruptive authorised correction and retest.	Successful event with new trace or safe replay.	Preserve before/after evidence.
6	Escalate security or data-integrity events using the incident SOP.	Incident owner and record.	Do not delete logs or receipts.

15.2 Diagnostic Order

Order	Check	Evidence
1	Correct plane, hostname and proxy route	421/404/302 and proxy access log
2	Service process and health	Master/Agent health and service manager
3	User/client active, role, API key and session	Registry/account state
4	Source IP/VPN and certificate/public key	Allowlist and fingerprint
5	Query/write/document mapping and permission	Registry and database grant
6	Agent route, transit secret and target profile	Agent log and exact endpoint status
7	Database objects, owner, function and grants	Verification report/pg_catalog
8	Business contract/parameters/versioning	Response code and approved definition
9	Timeout/idempotency receipt	Trace plus receipt table

16 AIra Assistant

16.1 Ask AIra

Use the assistant for DOORS knowledge or authorised activity questions.

Step	What to do	What you should see	Important control or tip
1	Open AIra and ask one specific question in plain language.	Answer with relevant source/operational basis.	Examples: explain rate limiting; steps to create a write service.
2	For live counts, state the time scope and activity, such as documents downloaded today.	Authorised live result or clear inability.	Static documentation must not invent current counts.
3	Review cited source and distinguish guidance from current system state.	Relevant source title/data source.	Unrelated topics should not be appended.
4	Rate the answer or report a gap where available.	Feedback recorded.	Do not include passwords or sensitive payloads in questions.

16.2 Reindex Knowledge

Only an authorised administrator reindexes AIra after an approved knowledge-source change. The administrator records source version/checksum, invokes the protected reindex operation with an authenticated session and CSRF token, then tests feature, workflow, live-activity, source-attribution and unauthorised-scope questions. Reindexing is not required after an ordinary user question.

17 User Security and Platform Administration

17.1 Register or Approve a User

Create the minimum necessary human access.

Step	What to do	What you should see	Important control or tip
1	Obtain official identity, organisation, required role and business justification.	Approved source request.	Do not create shared accounts.
2	Open user administration and create/review the account.	Pending user record.	Username/email must be unique.
3	Assign the minimum role, Agent/client scope, VPN requirement and allowed IPs.	Explicit entitlements.	Identity administration does not grant data approval automatically.
4	Approve/activate and provide temporary onboarding through the approved channel.	Active account with reset/MFA requirement.	Do not transmit credentials together with all recovery information.
5	Verify first login and audit events.	Password change, MFA and login evidence.	Close onboarding only after verification.

17.2 Change or Remove Access

Handle transfer, suspension or separation.

Step	What to do	What you should see	Important control or tip
1	Receive approved request with effective time and reason.	Change reference.	Urgent compromise suspension may precede normal closure paperwork.
2	Identify roles, IP/VPN, Agents, clients and owned pending work.	Complete impact list.	Prevent orphan approvals/services.
3	Reassign accountable work where required.	New owner with audit event.	Original proposer/reviewer history remains.
4	Disable/change account and revoke sessions/MFA challenges.	New login denied or reduced role active.	Do not wait only for browser expiry.
5	Review access used after effective time and close evidence.	Deprovisioning record.	Escalate anomalies.

17.3 Manage Categories and Subcategories

Maintain controlled classification used by requests and services.

Step	What to do	What you should see	Important control or tip
1	Open Category Master with an authorised administrator role.	Category list and state.	Search before creating a near-duplicate.
2	Create or edit the display name, code, description and parent relationship.	Validated category record.	Codes should remain stable after use.
3	Activate only after confirming affected forms and reporting.	Selectable category.	Activation does not approve a service.
4	Disable unused values after impact review.	Hidden for new records but retained historically.	Do not delete values referenced by audit history.

17.4 Manage Template Contracts

Publish reusable structures while retaining version control.

Step	What to do	What you should see	Important control or tip
1	Open Template Contracts and create or revise a template.	Draft template and version.	State the intended request/service type.
2	Define required fields, descriptions, validation and examples.	Previewable structure.	Examples must not contain live secrets or personal data.
3	Review independently and publish the approved version.	Active version and history.	Existing submissions remain tied to their version.
4	Deprecate or replace through a governed revision.	Successor and retirement state.	Do not silently change meaning for existing users.

17.5 Manage PKI Certificates

Register and rotate public trust material without receiving client private keys.

Step	What to do	What you should see	Important control or tip
1	Open PKI Certificate Management and select the owner/client/Agent.	Certificate inventory.	Use the correct environment.
2	Upload or paste only the public certificate and analyse it.	Subject, issuer, serial, fingerprint and validity.	Never upload a Production private key or password.
3	Compare the fingerprint through an independent channel and approve/activate.	Trusted active certificate.	A configured certificate does not grant service access.
4	Schedule rotation before expiry and support controlled overlap where policy allows.	New certificate active with recorded dates.	Revoke compromised material immediately.
5	Retest signed/encrypted API flows and preserve evidence.	Successful verification and trace.	Remove old trust only after consumers migrate.

18 Routine Operations and Release Checks

18.1 Daily Data Manager Checklist

Check	Expected condition	Action if not
Pending governance queues	Known owners and acceptable age	Assign/review/escalate
Agent readiness	Expected Agents healthy; no unrelated rows spinning	Inspect affected Agent only
Failed/denied requests	Known cause or active investigation	Use trace and classify
Version-limit/rate-limit events	Within expected policy	Contact owner; investigate loops
Document failures/checksum mismatch	Zero unexplained integrity failures	Security/operations escalation
Client/key/certificate expiry	No unexpected imminent expiry	Plan controlled rotation
Database verification	Mapped write Agents remain verified	Stop new activation and investigate drift

18.2 Release and Deployment Checklist

1. Confirm approved change scope, affected UI/Master/Agent/database components and rollback plan.
2. Build UI, Master and Agent independently; record SHA-256 checksums.
3. Back up binaries, external configuration and affected databases.
4. Apply Master Flyway and GePNIC service migrations in approved order.
5. Preserve environment-specific application.yml, secrets, truststores and identities.
6. Deploy UI under the correct base path and clear/invalidate stale browser/proxy caches.
7. Restart/reload through the approved service manager and verify routes, CSRF, health and TLS.
8. Run login, query, report, write, document, audit and AIra smoke tests plus unaffected-service regression.
9. Record acceptance or rollback decision and attach evidence.

19 Error Messages and Recovery

Status/code	Meaning	User action
Failed to load security token	Master unavailable, wrong route/base path, TLS/browser request failure	Check Master health and CSRF route; confirm local/staging UI build and proxy
401	Authentication/session/credential missing or expired	Sign in again or correct machine credentials
403/current IP not whitelisted	Role, IP, key, signature or service denial	Use trace and correct the specific policy
404	Route/resource absent or old deployment	Check proxy path and deployed Master/Agent build
409 WRITE_WORKFLOW_STATE	Prerequisite/state not satisfied or downstream controlled failure	Read response detail; complete mapping/test/provisioning
421	Wrong request plane/host	Use the assigned Admin, External or API hostname
422	Invalid parameters, JSON contract, Agent profile, schema or permission	Use exact field/Agent diagnostic and correct it
429	Rate limit exhausted	Wait and back off
500	Unexpected server failure	Record trace; check Master log; do not expose stack traces to user
502/503	Agent/dependency unavailable	Check service and connectivity; retry only when safe
504 WRITE_RESULT_UNKNOWN	Write outcome uncertain	Retry identical payload with same request ID

20 Role Checklists and Escalation

Role	Before task	Before declaring complete
External User	Correct plane, purpose and minimum data	Request ID and status visible
Developer	Approved need, SELECT-only SQL and :p_ inputs	Successful dry run and submitted proposal
Reviewer	Independent account and full definition	Current test, remarks and recorded decision
Data Viewer	Authorised report and values	Expected results/trace and safe export
Data Manager	Correct environment, client/service/Agent	All independent permissions and evidence confirmed
Security Administrator	Approved identity/access request	Least privilege, session revocation and audit history
DBA	Approved SQL/hash, backup and role inventory	Objects, owner, function settings, grants and verification pass
Application Owner	Checksum, wallet/trust, IP and public-key registration	UAT success, log evidence and protected Production identity
Operator	Approved build/checksum/config/rollback	Health and complete smoke/regression evidence
Auditor	Approved scope and read-only evidence access	Traceable exceptions, owners and management responses

20.1 Escalation Information

- Environment and exact URL without credentials.
- User role or client ID/name; include API key only through the explicitly approved protected troubleshooting channel.
- Date/time with timezone.
- Operation, service/query, Agent and business key where non-sensitive.
- HTTP status, application code, request ID and DOORS trace ID.
- Screenshot or log excerpt with passwords, private keys, wallet secrets and personal data removed.
- Whether retry was attempted and whether the payload/request ID changed.

Appendix A Field and Status Reference

Domain	Statuses	Meaning
User	Pending, Approved/Active, Rejected, Disabled	Account governance and admission
Request	Pending, Approved, Rejected/Returned	Business request decision
Query	Draft/Pending, Tested, Approved, Rejected, Disabled	Read definition lifecycle
Write proposal	Draft, Sandbox Tested, Pending Approval, Approved, Rejected, Disabled	Contract lifecycle
Agent mapping	Verification pending, Verified, Failed	Destination schema/readiness
API client	Inactive, Active	Machine admission; not a data permission
Write response	STORED, UNCHANGED, replayed, conflict, denied, unknown	Database business outcome

Appendix B Security Do and Do Not Guide

Do	Do not
Use your own named account and authorised role	Share portal accounts or approve using another person's session
Keep private keys and PKCS12 passwords with the application owner	Upload/email Production private keys to DOORS administrators
Use generated SDK encryption/signing and verify checksums	Write an unreviewed client that omits signing, encryption or trust validation
Use exact same request ID and payload after unknown write result	Reuse a request ID with changed JSON
Use generated DBA SQL and verify definition hash	Grant doors schema ownership, DDL or direct table write
Search logs using trace IDs and authorised readable context	Log passwords, private keys, AES session keys or raw authentication credentials
Create governed revisions for SQL/contract changes	Edit approved production definitions or database tables informally
Report certificate warnings and stop	Disable TLS verification in Production

Appendix C Glossary

Term	Meaning
Agent	DOORS service near one governed GePNIC database/document node
API key	Client passport identifier; not itself a data permission
Business or record key	Approved JSON fields identifying one versioned record
Contract version	Version of JSON structure, required fields, types, keys and projections
Definition hash	SHA-256 binding generated DBA SQL to approved contract
Idempotency	Exact retry does not create a duplicate business effect
Instance ID	Stored destination context available for consolidated/multi-instance use
Plane	Isolated Admin, External or API request surface
Record version	Automatically assigned immutable changed snapshot number
SDK	Generated client implementing approved endpoint and security protocol
Trace ID	Identifier correlating UI/gateway, Master, Agent and error evidence
TOTP	Time-based one-time password used for MFA